When calculating matrix elements

$$\int_0^{r=t_{\text{last}}} B_i^k(r) H B_j^k(r) dr \tag{1}$$

we need of course only to integrate over the knotpoints where both B_i and B_j are non-zero

$$\int_{r=t_{\max(i,j)}}^{r=t_{\min(i,j)+k}} B_i^k(r) H B_j^k(r) dr. \tag{2}$$

It is possible to use Gaussian integration directly here, but you might find it simpler to first divide the integration into several segments- one for each knotpoint:

$$\int_{r=t_{\max(i,j)}}^{r=t_{\min(i,j)+k}} B_i^k(r) H B_j^k(r) dr = \sum_{m=\max(i,j)}^{\min(i,j)+k-1} \int_{r=t_m}^{r=t_{m+1}} B_i^k(r) H B_j^k(r) dr \tag{3}$$

Now we may use Gaussian integration with say k abscissas in each interval (sine the B-splines are polynomials of order $k-1$, and the operators change this only with one or two steps, around k should be OK, but you can check!):

$$\sum_{m=\max(i,j)}^{\min(i,j)+k-1} \int_{r=t_m}^{r=t_{m+1}} B_i^k(r) H B_j^k(r) dr \approx \sum_{m=\max(i,j)}^{\min(i,j)+k-1} \sum_{n=1}^{k} w_n f(r_{n_m}) \tag{4}$$

The reason that this is simpler is that now we can just integrate segment by segment (in the outer loop) and do this for all B-splines which are non-zero in that segment (inner loop) . The results for each pair of B-splines is then just be summed into the matrix-element H_{ij}. But this can be solved in many ways so you can arrange it in a way that feels natural for you.

Remember that the abscissas have to be transformed to fit each interval:

$$\begin{aligned}
\int_a^b f(x)\, dx = & \\
& \left[z = \frac{x-a}{b-a} - \frac{b-x}{b-a} \rightarrow x = z\frac{b-a}{2} + \frac{a+b}{2},\ dx = \frac{b-a}{2} dz \right] \\
& = \frac{(b-a)}{2} \int_{-1}^{1} f(z\frac{b-a}{2} + \frac{a+b}{2})\, dz \approx \frac{(b-a)}{2} \sum_{i=1}^{n} w_i f(z_i \frac{b-a}{2} + \frac{a+b}{2})
\end{aligned} \tag{5}$$